Introduction to Logic (CS) 2026

Week 1, Lecture 2

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Thursday 3 Sep 2026

1. Course organization
2. Formal Proofs with Atomic Sentences
3. Boolean connectives and compound sentences
4. Proof with Boolean connectives

Today's Lecture

1. Course organization
2. Formal Proofs with Atomic Sentences
3. Boolean connectives and compound sentences
4. Proof with Boolean connectives

Tutorials

Did everyone manage to find their tutorial group? Have you tried Fitch proofs at the tutorial?

Normal setup

Setup most of the time:

- Lectures: Monday/Tuesday
- Tutorials: Wed/Thursday

Assessment

Assessment components:

- Two homework assignments (HW1, HW2)

Homework to be submitted in *Brightspace* and in *Themis*.
Submissions in the PDF format: either generate the document digitally, or use pen and paper + neatly scan it.

Assessment

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- Midterm test (M)

Assessment

Assessment components:

- Two homework assignments (HW1, HW2)
- Midterm test (M)
- Exam (E), Final Grade (G)

To pass the course you have to get at least 5.0 on your exam.

- E higher than HW,M $\Longrightarrow$ G = E;
- otherwise $\Longrightarrow$ G = (8$*$HW1+16$*$HW2+16$*$M+60$*$E)/100;
- if you take a resit (R) $\Longrightarrow$ G = R;
- G $\geq$ 6.0 is needed to pass. (Note: 5.5 rounds to 6.0 and 5.4 rounds to 5.0.)

Today's Lecture

1 Course organization

2 Formal Proofs with Atomic Sentences

3 Boolean connectives and compound sentences

4 Proof with Boolean connectives

Atomic sentences

So far we saw *atomic sentences* made out of

- constants (*a*, *b*, *Alex*...)

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- constants (a, b, *Alex*...)
- function applications ($f(x)$, $g(h(a,b), n + (m - k)$, ...)

Atomic sentences

So far we saw *atomic sentences* made out of

- constants (a, b, *Alex*...)
- function applications ($f(x)$, $g(h(a,b), n + (m - k)$, ...)
- propositions and predicates (A, B, $P(a)$, $Q(f(a), g(b))$, ...)
- equality ($a = b$, $f(a) = h(g(a), b)$, ...)

To show validity of an argument, give a proof!

From Language, Proof and Logic p. 46-47:

> *A proof is a step-by-step demonstration that a conclusion follows from some premises...*

General shape of proofs in system $\mathcal{F}$

1.	P_1	(Premise)
2.	P_2	(Premise)
3.	P_3	(Premise)
4.	T_1	Justification
5.	T_2	Justification
⋮		⋮
k.	C	Justification

...possibly with nested subproofs.

- Rules: introduction and elimination of logic constructs; reiteration.
- Justification: how a sentence follows from earlier ones by applying the rules.
- **Today's rules:** =Introduction, =Elimination, Reiteration

= Introduction (system $\mathcal{F}$ rule)

Idea: Every object is identical to itself.
Example: Zorro is the same person as Zorro.

I.e., for all terms (incl. names) t, we have that $t = t$. Identity is reflexive.

= Introduction (system $\mathcal{F}$ rule)

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General form:

$$\begin{array}{|l} \vdots \\ \text{k.} \quad t = t \quad \text{=Intro} \\ \vdots \end{array}$$

Examples:

$$\begin{array}{|lll} \vdots & & \\ 5. & \mathit{zorro} = \mathit{zorro} & \text{=Intro} \\ 6. & \mathit{father}(\mathit{claire}) = \mathit{father}(\mathit{claire}) & \text{=Intro} \\ \vdots & & \end{array}$$

= Elimination (system $\mathcal{F}$ rule)

Idea:
For all terms s and t, if $s = t$ then you may replace any occurrence of s with t in any sentence.

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Idea:
For all terms *s* and *t*, if $s = t$ then you may replace any occurrence of *s* with *t* in any sentence.

Examples:

Consider

constant	ddlv:	Don Diego de la Vega
constant	adlv:	Don Alejandro de la Vega
constant	zorro:	Zorro
unary predicate	C(x):	x is from California

1. ddlv = zorro
2. C(ddlv)

3. C(zorro) =Elim: 2, 1

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2. C(ddlv)

---

3. C(zorro) =Elim: 2, 1

1. C(adlv)
2. adlv = father(ddlv)
3. ddlv = zorro

---

4. C(father(ddlv)) =Elim: 1, 2
5. C(father(zorro)) =Elim: 4, 3

= Elimination (system $\mathcal{F}$ rule)

Idea:
For all terms s and t, if $s = t$ then you may replace any occurrence of s with t in any sentence.

General form:

⋮		
i.	$A(s)$	Justification (or premise)
⋮		
j.	$s = t$	Justification (or premise)
⋮		
k.	$A(t)$	= Elim: i, j
⋮		

Note:

- $A(s)$ is a pattern. It denotes a sentence in which s occurs.
- The order in which $A(s)$ and $s = t$ occur above the application of =Elim does not matter.
- The order of the references i,j in =Elim: i,j does matter! First reference to the pattern $A(s)$. Then reference to identity $s = t$.

Example: Proof of transitivity of =

Identity is transitive:

For all terms s, t, u: if $s = t$ and $t = u$ then $s = u$.

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Identity is transitive:

For all terms s, t, u: if $s = t$ and $t = u$ then $s = u$.

Proof in system $\mathcal{F}$:

1. $s = t$
2. $t = u$
3. $s = u$ =Elim: 1, 2 *(in line 1, replace t with u)*

Note again the order of references in the justification:
First the pattern, then the identity.
Why is =Elim: 2, 1 incorrect?

Example: Proof of symmetry of =

Identity is symmetric:

For all terms s and t:

if $s = t$
then $t = s$.

In Fitch form:

$s = t$

$t = s$

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Proof in system $\mathcal{F}$:

1.	$s = t$		
2.	$s = s$	=Intro	
3.	$t = s$	=Elim: 2, 1	*(in line 2, replace the left s with t)*

Note: the order of justification references:
First the pattern, then the identity.
In line 3: = Elim: 1, 2 would not be correct (why?).

Properties of identity

We have formally proved in system $\mathcal{F}$ that the identity relation = has the properties:

reflexivity: For all t: $t = t$

symmetry: For all s, t: if $s = t$ then $t = s$

transitivity: For all s, t, u: if $s = t$ and $t = u$ then $s = u$

A problem?

? How to prove:

1. $P(a)$
2. $b = a$

---

3. $P(b)$

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3. $P(b)$ =Elim: 1, 2 **??**

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3. $P(b)$ =Elim: 1, 2 **??**

Not allowed in $\mathcal{F}$: $b = a$ only allows us to replace b by a, not the other way round.

Solution

? How to prove:

1. $P(a)$
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---

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Solution:

1. $P(a)$
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3. $b = b$ =Intro

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Solution:

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3. $b = b$ =Intro
4. $a = b$ =Elim: 3, 2

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Solution:

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4. $a = b$ =Elim: 3, 2
5. $P(b)$ =Elim: 1, 4

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Solution:

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3. $b = b$ =Intro
4. $a = b$ =Elim: 3, 2
5. $P(b)$ =Elim: 1, 4

Prove an instance of symmetry on the way.

Replacing one or more occurrences

1. $Q(a, a, b, b)$
2. $a = c$
3. $b = d$

---

4. $Q(c, a, b, b)$ =Elim: 1, 2
5. $Q(c, a, d, d)$ =Elim: 4, 3

Replacing one or more occurrences

1.	$Q(a, a, b, b)$	
2.	$a = c$	
3.	$b = d$	
4.	$Q(c, a, b, b)$	=Elim: 1, 2
5.	$Q(c, a, d, d)$	=Elim: 4, 3

In step 4, one occurrence of a is replaced by c

In step 5, two occurrences of b are replaced by d

System $\mathcal{F}$ rules: Reiteration

Evidently,

No human is a daffodil.

...

No human is a daffodil.

System $\mathcal{F}$ rules: Reiteration

Evidently,

> No human is a daffodil.
>
> ...
>
> No human is a daffodil.

Reiteration rule allows to restate sentences that have already been proved:

⋮		
j.	A	Justification (or premise)
⋮		
k.	A	Reit: j
⋮		

When proofs get very long, the Reiteration rule can be convenient.
But it is also crucial for compound sentences.

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Atomic and compound sentences

Recall:

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- An atomic sentence is a sentence that cannot be broken down further
- How do we make sentences beyond atomic ones?

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Starting points of Boolean logic:

- There are precisely two truth-values: $\{\mathsf{T}, \mathsf{F}\}$
- Everything is either true or false

Atomic and compound sentences

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- A sentence is a statement that either True or False
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Starting points of Boolean logic:

- There are precisely two truth-values: $\{\mathsf{T}, \mathsf{F}\}$
- Everything is either true or false

This is not true in natural language!

- *This sentence is false*

Liar's paradox

This sentence is false,

The next sentence is true. The previous sentence is false.

There are many non-Boolean logics that make different assumption (e.g. three-valued logics, probabilistic logics, etc); outside of the scope of this class.

History
Explanation and variants
Possible resolutions
Fuzzy logic
Alfred Tarski
Arthur Prior
Saul Kripke
Jon Barwise and John Etchemendy
Dialetheism
Non-cognitivism
Bhartrhari's perspectivism
Logical structure
Applications
Gödel's first incompleteness theorem
In popular culture
See also

Making compound sentences

Atomic sentences:

- You are attending this lecture in person
- The train arrives on time

Compound sentences:

Making compound sentences

Atomic sentences:

- You are attending this lecture in person
- The train arrives on time

Compound sentences:

- You are not attending this lecture in person
- You are attending this lecture in person, and the train arrives on time
- You are either attending this lecture in person, or the train arrives on time
- You are attending this lecture and the train does not arrive on time

Compositionality of truth

Is “You are attending this lecture in person, and the train arrives on time” true?

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- If both sub-sentences are true, then the compound sentence is true. False otherwise.

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- Split into: “You are attending this lecture in person” & “The train arrives on time”
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- Is “The train arrives on time” true?
- If both sub-sentences are true, then the compound sentence is true. False otherwise.

Compositionality of truth: The truth value of the compound sentence is determined by the truth value of its parts.

Negation

"You are not attending this lecture in person"

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$$\neg P$$

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"You are not attending this lecture in person"

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If P is true, then $\neg P$ is false.
If P is false, then $\neg P$ is true.

Negation

"You are not attending this lecture in person"

$$\neg P$$

If P is true, then $\neg P$ is false.
If P is false, then $\neg P$ is true.

P	$\neg P$
T	F
F	T

N.b. we write $a \neq b$ for $\neg(a = b)$. How do you write negation in C?

Conjunction

"You are attending this lecture in person, and the train arrives on time"

$$P \wedge Q$$

- If P is true and Q is true, then $P \wedge Q$ is true.
- If P is true and Q is false, then $P \wedge Q$ is false.
- If P is false and Q is true, then $P \wedge Q$ is false.
- If P is false and Q is false, then $P \wedge Q$ is false.

Conjunction

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P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Translating from natural language (1)

Q: How to split the sentences in these examples?

- Mary came in and started to laugh.

Translating from natural language (1)

Q: How to split the sentences in these examples?

- Mary came in and started to laugh.
  - *Incorrect: P* : Mary came, *Q* : started to laugh;
  - *Correct: P* : Mary came, *Q* : Mary started to laugh;

  Subsentences have to make sense on their own

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- I know both Mary and John

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  - *Correct:* P : Mary came, Q : Mary started to laugh;

  Subsentences have to make sense on their own

- I know both Mary and John
  - I know Mary and I know John

NB: Non-commutativity of “and”

Sometimes “and” in a sentence presents more information than just a conjunction

Consider:

- *Kyle left and Chun started crying*; vs
- *Chun started crying and Kyle left*

Disjunction

"He is distracted or he didn't hear it"

$$P \vee Q$$

- If P is true, then $P \vee Q$ is true.
- If Q is true, then $P \vee Q$ is true.
- If P is false and Q is false, then $P \vee Q$ is false.

Disjunction

"He is distracted or he didn't hear it"

$$P \vee Q$$

- If P is true, then $P \vee Q$ is true.
- If Q is true, then $P \vee Q$ is true.
- If P is false and Q is false, then $P \vee Q$ is false.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Translating from natural language (2)

Q: Which connectives you can use to translate the following sentences?

1. John is smart and eager to please.

Translating from natural language (2)

Q: Which connectives you can use to translate the following sentences?

1. John is smart and eager to please.
2. Either Jake or Josh were there.

Translating from natural language (2)

Q: Which connectives you can use to translate the following sentences?

1. John is smart and eager to please.
2. Either Jake or Josh were there.
3. He tried to catch the bus, but it was too late.

Translating from natural language (2)

Q: Which connectives you can use to translate the following sentences?

1. John is smart and eager to please.
2. Either Jake or Josh were there.
3. He tried to catch the bus, but it was too late.
4. The weather was fine; moreover, the food was very good.

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1. John is smart and eager to please.
2. Either Jake or Josh were there.
3. He tried to catch the bus, but it was too late.
4. The weather was fine; moreover, the food was very good.
5. Although it was not a holiday, the shop was closed.

Translating from natural language (2)

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1. John is smart and eager to please.
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5. Although it was not a holiday, the shop was closed.
6. We arrived early; however there was no place left.

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A: Conjunction for 1,3-6, disjunction for 2, negation for 6

Translating from natural language (3)

Q: Which connectives you can use to translate the following sentences?

1. Neither the participants nor the organizers had foreseen this.
2. The participants nor the organizers had forseen this.
3. Neither Josh nor Jake were there.

Translating from natural language (3)

Q: Which connectives you can use to translate the following sentences?

1. Neither the participants nor the organizers had foreseen this.
2. The participants nor the organizers had forseen this.
3. Neither Josh nor Jake were there.

- It is not the case that either Josh or Jake were there.
- Josh was not there and Jake was not there.

Translating from natural language (3)

Q: Which connectives you can use to translate the following sentences?

1. Neither the participants nor the organizers had foreseen this.
2. The participants nor the organizers had forseen this.
3. Neither Josh nor Jake were there.

- It is not the case that either Josh or Jake were there.
- Josh was not there and Jake was not there.

A: neither A nor B is translated as $\neg(A \vee B)$ or $\neg A \wedge \neg B$

NB: Exclusive and inclusive “or”

Inclusive disjunction:

- He is distracted or he didn’t hear it

Exclusive disjunction (`xor`):

- He is either in England or in France [but not in both places]

How do you translate these sentences?

NB: Exclusive and inclusive “or”

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Exclusive disjunction (`xor`):

- He is either in England or in France [but not in both places]

How do you translate these sentences?

A xor B is translated as $(A \wedge \neg B) \vee (\neg A \wedge B)$, or $(A \vee B) \wedge \neg(A \wedge B)$

Example of a complete translation

Barbie either stays in Barbieland and does not live an authentic life, or she goes on an adventure and discovers her true self.

Translation key:

Translation:

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Barbie either stays in Barbieland and does not live an authentic life, or she goes on an adventure and discovers her true self.

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- P: Barbie stays in Barbieland.
- Q: Barbie lives an authentic life.
- R: Barbie goes on an adventure.
- S: Barbie discovers her true self.

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Example of a complete translation

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- P: Barbie stays in Barbieland.
- Q: Barbie lives an authentic life.
- R: Barbie goes on an adventure.
- S: Barbie discovers her true self.

Translation: $(P \wedge \neg Q) \vee (R \wedge S)$

Translating from natural language (4)

Q: Which connectives you can use to translate the following sentences?

1. You will miss the train unless you start packing.
2. Unless the flag is set, the program will crash.

Translating from natural language (4)

Q: Which connectives you can use to translate the following sentences?

1. You will miss the train unless you start packing.
2. Unless the flag is set, the program will crash.
3. You either start packing or you will miss the train
4. The flag either is set or the program will crash.

Translating from natural language (4)

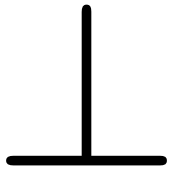
Q: Which connectives you can use to translate the following sentences?

1. You will miss the train unless you start packing.
2. Unless the flag is set, the program will crash.
3. You either start packing or you will miss the train
4. The flag either is set or the program will crash.

A: P unless Q is translated as $Q \vee P$
N.b. we will see another possible translation of unless next week

Falsity

"False"



$\bot$ is an (atomic) sentence
that is always false

N.b. we define $\top = \neg\bot$

The language of Boolean logic

Given a collection of atomic sentences (e.g. atomic propostitions or FOL atomic sentences), a *formula* is constructed as follows:

- An atomic sentence P is a formula.
- $\bot$ is a formula.
- If P is a formula, then $\neg P$ is a formula.
- If P and Q are formulas, then $(P \wedge Q)$ is a formula.
- If P and Q are formulas, then $(P \vee Q)$ is a formula.

The language of Boolean logic

Given a collection of atomic sentences (e.g. atomic propostitions or FOL atomic sentences), a *formula* is constructed as follows:

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- If P and Q are formulas, then $(P \wedge Q)$ is a formula.
- If P and Q are formulas, then $(P \vee Q)$ is a formula.

Examples: $\neg\neg A$, $(A \wedge \neg B) \vee C$, $\neg(\neg A \vee \neg B) \wedge \neg C$, etc.

Ambiguity in formulas

If A and B are false and C is true, what is the truth value of

$$A \wedge B \vee C$$

Ambiguity in formulas

If A and B are false and C is true, what is the truth value of

$$A \wedge B \vee C$$

- Ambiguity can be dangerous: $(A \wedge B) \vee C$ and $A \wedge (B \vee C)$ result in different truth values.
- Use parenthesis to disambiguate: *never write* $A \wedge B \vee C$.
- No need to use parenthesis when its never ambiguous: $A \wedge (B \wedge C)$ is the same as $(A \wedge B) \wedge C$, so we write it as $A \wedge B \wedge C$
- Similarly for $A \vee B \vee C$
- N.b. we write $\neg\neg P$ for $\neg(\neg P)$ and $a \neq b$ for $\neg(a = b)$

Some common issues with translations

Boolean connectives operate on Boolean values, i.e. you can take a conjunction of two formulas/sentences, but not of two terms!

Q: How do you translate to FOL: *Anna is friends with Bob and Clair*?

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Q: How do you translate to FOL: *Anna is friends with Bob and Clair*?
Bad **A:** $\mathit{Friends}(a, b \wedge c)$

Some common issues with translations

Boolean connectives operate on Boolean values, i.e. you can take a conjunction of two formulas/sentences, but not of two terms!

Q: How do you translate to FOL: *Anna is friends with Bob and Clair*?
Bad **A:** $\mathit{Friends}(a, b \wedge c)$ $b \wedge c$ does not make sense
Good **A:** $\mathit{Friends}(a, b) \wedge \mathit{Friends}(a, c)$

Today's Lecture

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3. Boolean connectives and compound sentences
4. Proof with Boolean connectives

Proof of conjunction

Claim: $P \wedge Q$

Proof method: Prove both P and Q

Proof of conjunction

Claim: $P \wedge Q$
Proof method: Prove both P and Q

Claim: I am teaching a lecture and it is sunny outside.
Proof: Go to the lecture room and see me in front of the presenating waving my hands around and talking about logic. This confirms that I am teaching a lecture. Separately look outside the window and examine the rays of sunshine. This confirms that it is sunny outside. Combining those two facts together, we obtain that I am teaching a lecture and it is sunny outside.

Proof of conjunction

...

...

...

i. P

...

j. Q

...

k. $P \wedge Q$ $\wedge$ Intro: i, j

Proof of conjunction

...

...

...

i_1. P_1

...

i_2. P_2

...

...

i_n. P_n

...

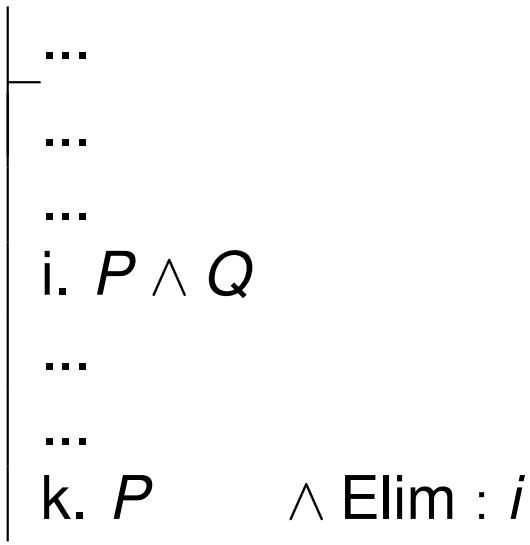
k. $P_1 \wedge P_2 \wedge \cdots \wedge P_n$ $\wedge$Intro : $i_1, i_2, \ldots, i_n$

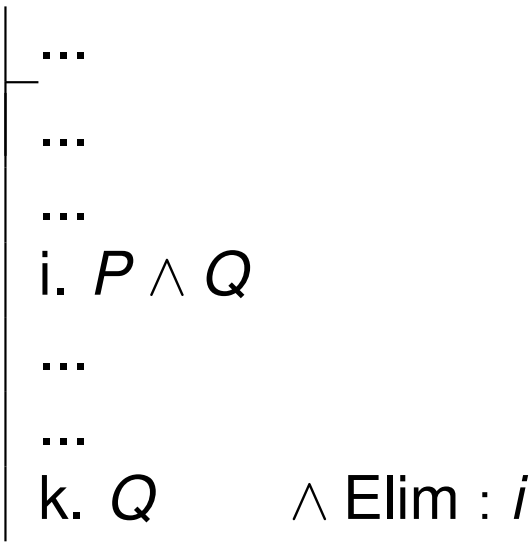
Elimination of conjunction

Conjunction elimination: From $P \wedge Q$ we can deduce P, and we can deduce Q.

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Example proofs with conjunction

Commutativity: $P \wedge Q \vdash Q \wedge P$

Associativity: $P \wedge (Q \wedge R) \vdash (P \wedge Q) \wedge R$

Proof of negation

Claim: $\neg P$

Proof method: Suppose that P holds, and arrive at a contradiction.

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Claim: Femke did not eat her sandwitch.
Proof: Suppose that Femke did in fact ate her sandwich. But if we open the fridge, we can see the sandwich right there. "Sandwich was eaten" $\wedge$ "sandwich is in the fridge" is a contradiction. Hence the original assumption that Femke ate her sandwich must be false.

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Proof: Suppose that $\sqrt{2}$ is rational.

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$$(\sqrt{2})^2 = 2 = (\frac{a}{b})^2 = \frac{a^2}{b^2}, \quad \text{and} \quad 2b^2 = a^2.$$

Hence a^2 and a are even numbers. Let us write $a = 2c$.

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Hence a^2 and a are even numbers. Let us write $a = 2c$.
Then $2b^2 = (2c)^2 = 4c^2$. Hence b^2 and b are both even. Let us write $b = 2d$. Then the fraction $\frac{a}{b} = \frac{2c}{2d}$ is reducible! Something cannot be both reduced and reducible further, hence we arrived at a contradiction. It means that our original assumption about $\sqrt{2}$ being rational was false. Hence, $\sqrt{2}$ is irrational.

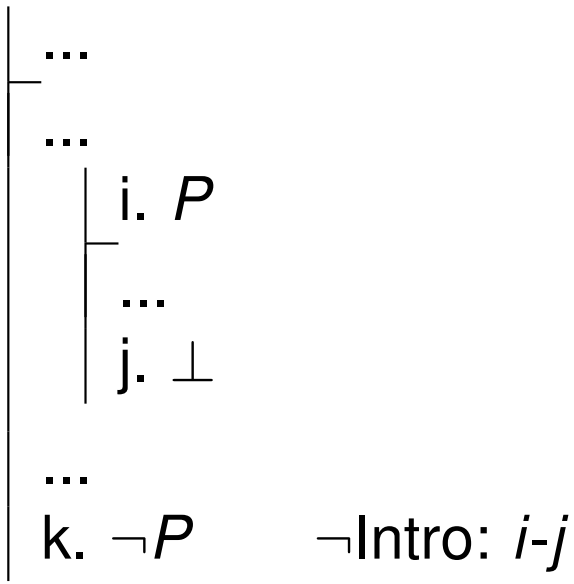
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Proof of negation



- The lines from *i* to *j* denote a *subproof*
- Subproofs can be nested inside each other

Example proof of negation

Not false is true:

$$\vdash \neg\bot$$

From P, deduce P and true:

$$P \vdash P \wedge \neg\bot$$

If P is true, then it's not the case that P is false:

$$P \vdash \neg\neg P$$

Elimination of negation

Double negation elimination:

| ...
|—
| ...
| ...
| i. $\neg\neg P$
| ...
| ...
| j. P $\neg$Elim: i

Contradiction rules

For proving $\neg P$ we need to arrive at a contradiction $\bot$... but how?

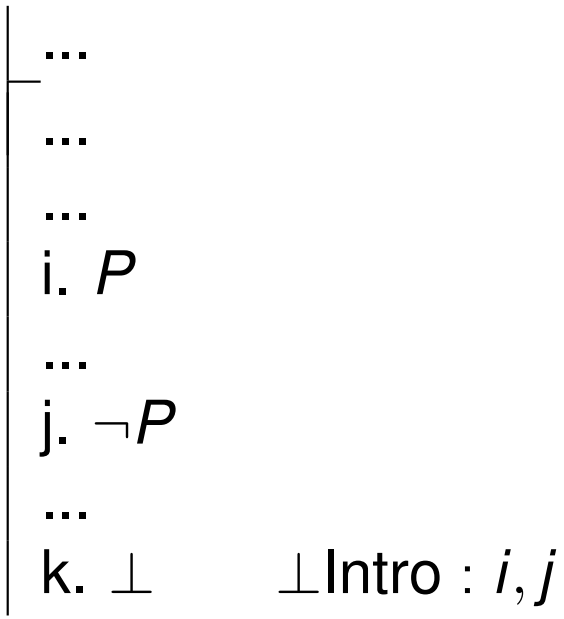
Contradiction rules

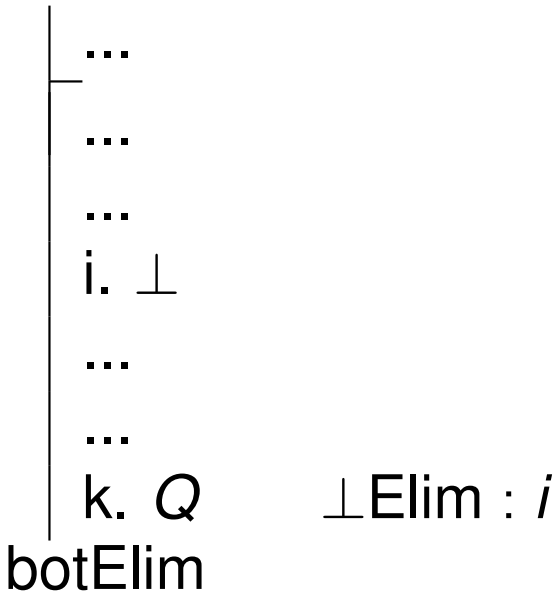
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| ...
|—
| ...
| ...
| i. P
| ...
| j. $\neg P$
| ...
| k. $\bot$ $\bot$Intro : i, j

Contradiction rules

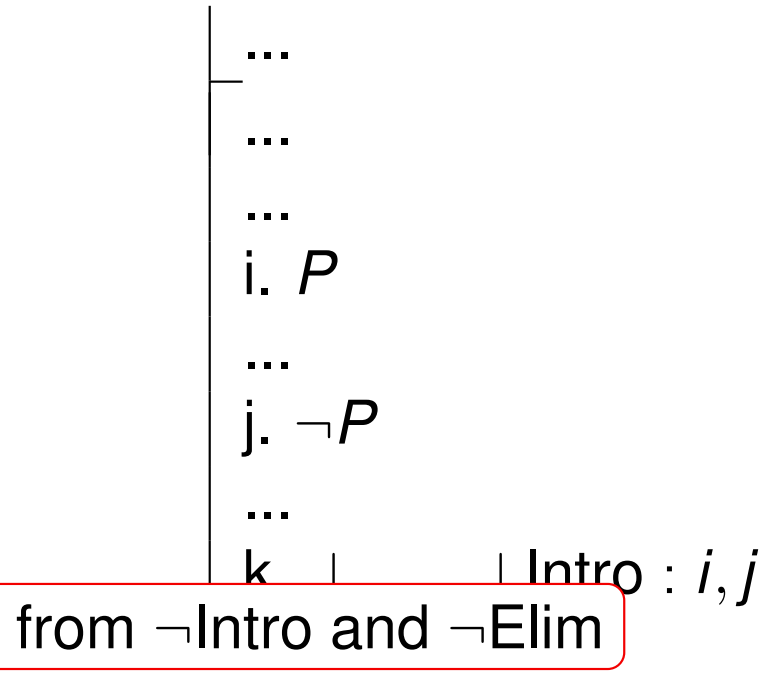
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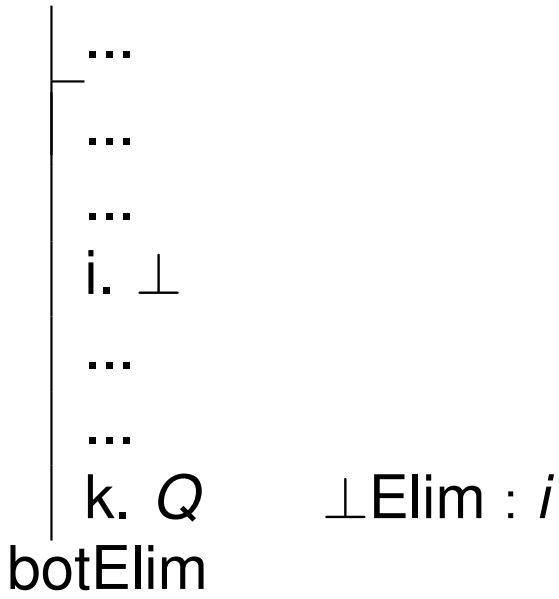




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Disjunction introduction

Claim: $P \vee Q$
Proof method 1: Prove P
Proof method 2: Prove Q

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...	...
...	...
...	...
i. P	i. Q
...	...
k. $P \vee Q$ $\vee$ Intro: i	k. $P \vee Q$ $\vee$ Intro: i